ETH zürich

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

# CachePool: Many-core cluster of customizable, lightweight scalar-vector PEs for irregular L2 data-plane workloads


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Open Source Hardware, the way it should be!

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# L1 Access Latency

- **Problem: increased L1 access latency when scaling up**
- **Solutions**
  - **(Ongoing, RTL & Physical)** Further optimize the interconnection and cache controller to reduce access latency on existing architecture
    => Retiming the current interconnection, remove unnecessary cuts in interco and cache
    => Target to reduce 1-2 cycles (~5 cyc) for hit-to-use latency
  - **(Planned, RTL)** Partition the shared L1 to keep data localized to avoid remote access latency
  - **(Ongoing, RTL & Physical)** Explore and evaluate a private WT L1 for each core
    => Ongoing master thesis, now is working on the coherence between L1 caches
    => Currently plan to use hierarchical directory-based approach

# L1 Access Latency

- **Problem: large NoC latency (~28 cyc based on previous exploration)**
- **Solutions**
  - **(Planned, Physical)** Explore the timing under 7nm technology to see the possibility of reducing per-hop latency to 1 cycle
    => Reduce the latency by 1/2
  - **(Planned, Physical)** If 1-cyc is not possible, explore to operate the NoC under a higher frequency than cores (e.g. 1.5GHz)
    => Effectively cut the latency by 1/3
  - **(Planned, GVSoC & Physical)** Explore different NoC topologies
    => Torus to reduce the diameter of NoC by 1/2, reduce longest latency by 1/2

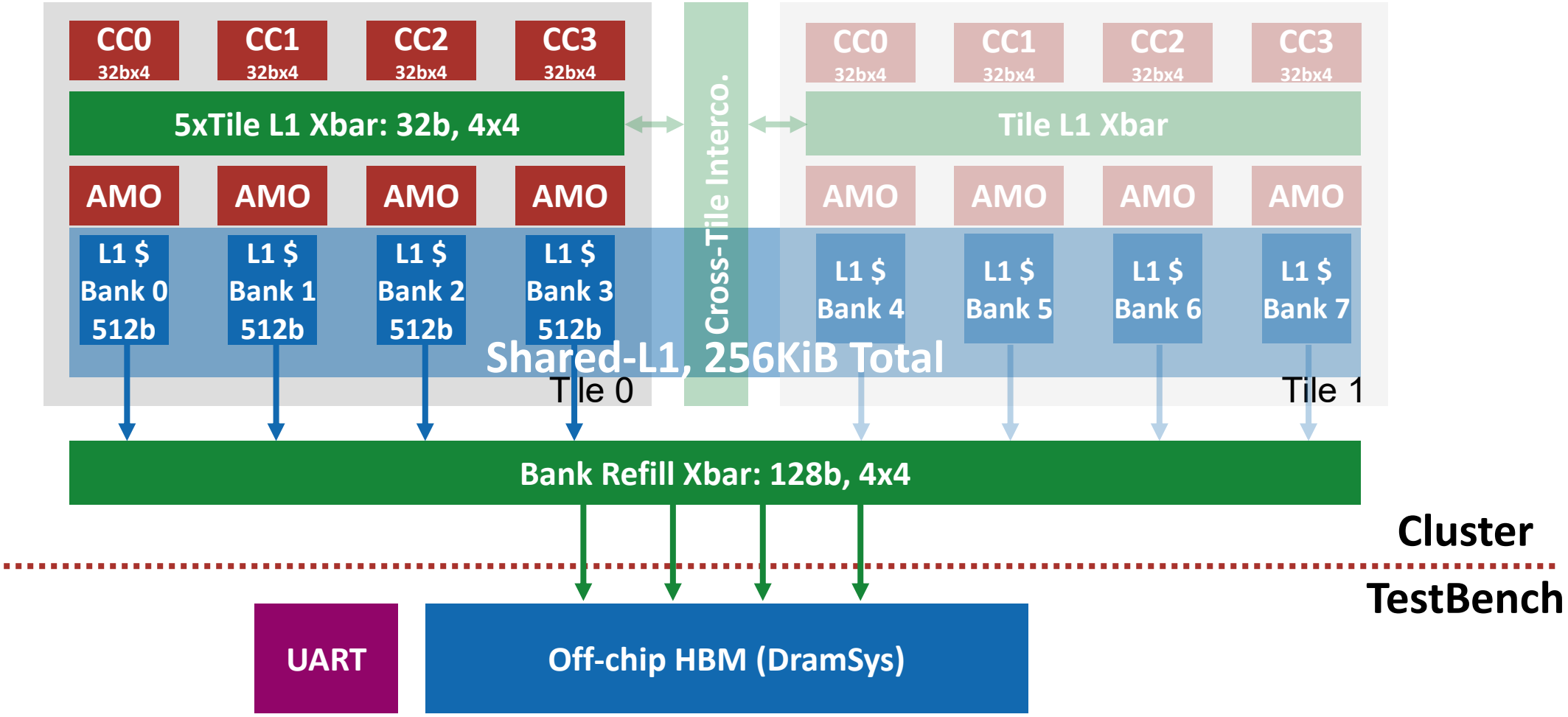
# PE-Related

- **Problem: Improve scalar core performance**
- **Solutions**
  - **(Planned, RTL & GVSoC)** Explore two Snitch + one Spatz configuration to improve scalar performance
  - **(Ongoing, RTL)** Change the host core (Snitch) to a more powerful scalar unit. This will be explored by a semester project in Feb.

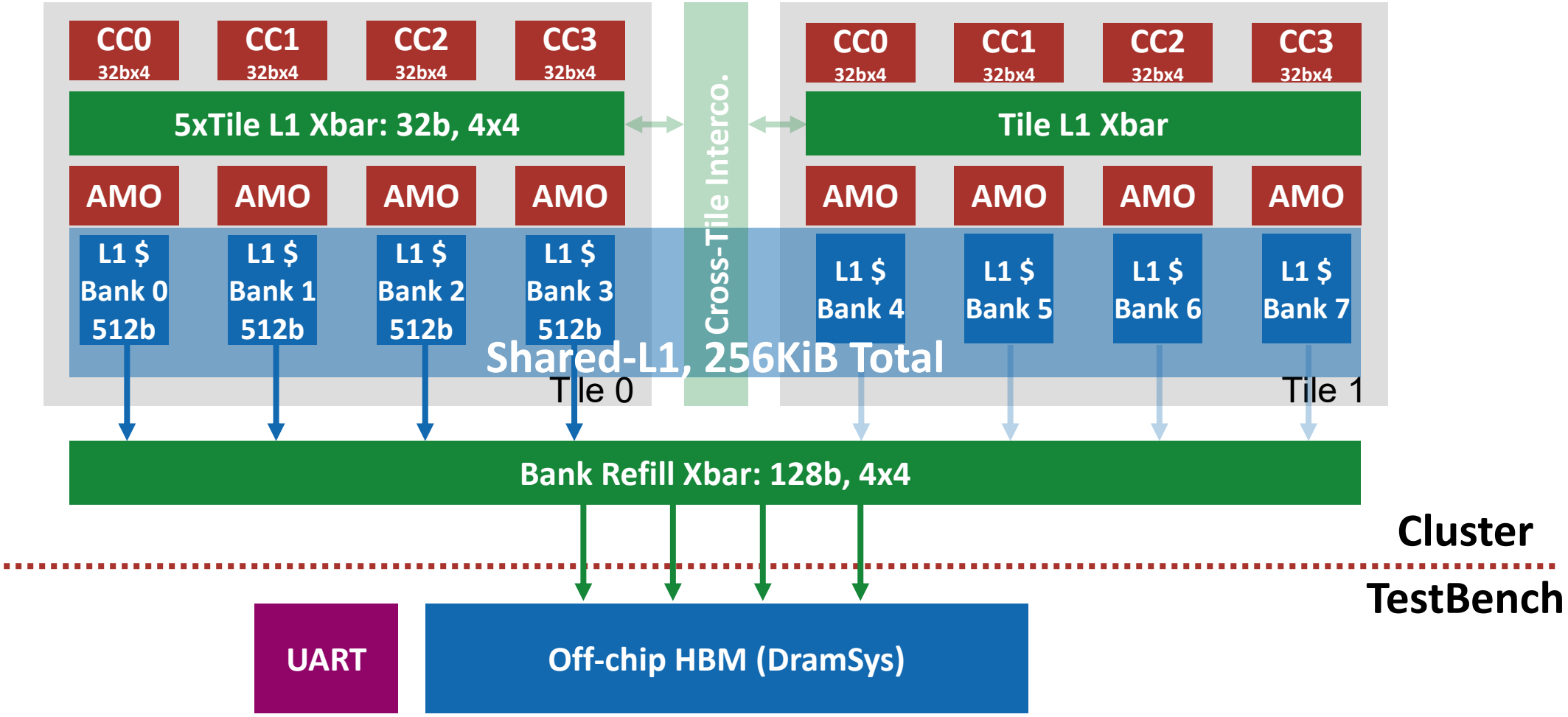
# RLC Workload

- **Problem: Current RLC model cannot reveal all properties of the kernel**
- **Solutions**
  - **(Planned, SW)** Adapt the current model to better represent the real cases, e.g. do some operations on the data
  - **(Planned, SW & GVSoC)** Further cooperation with HQ side to develop a more complete and better model.

# Hardware Development



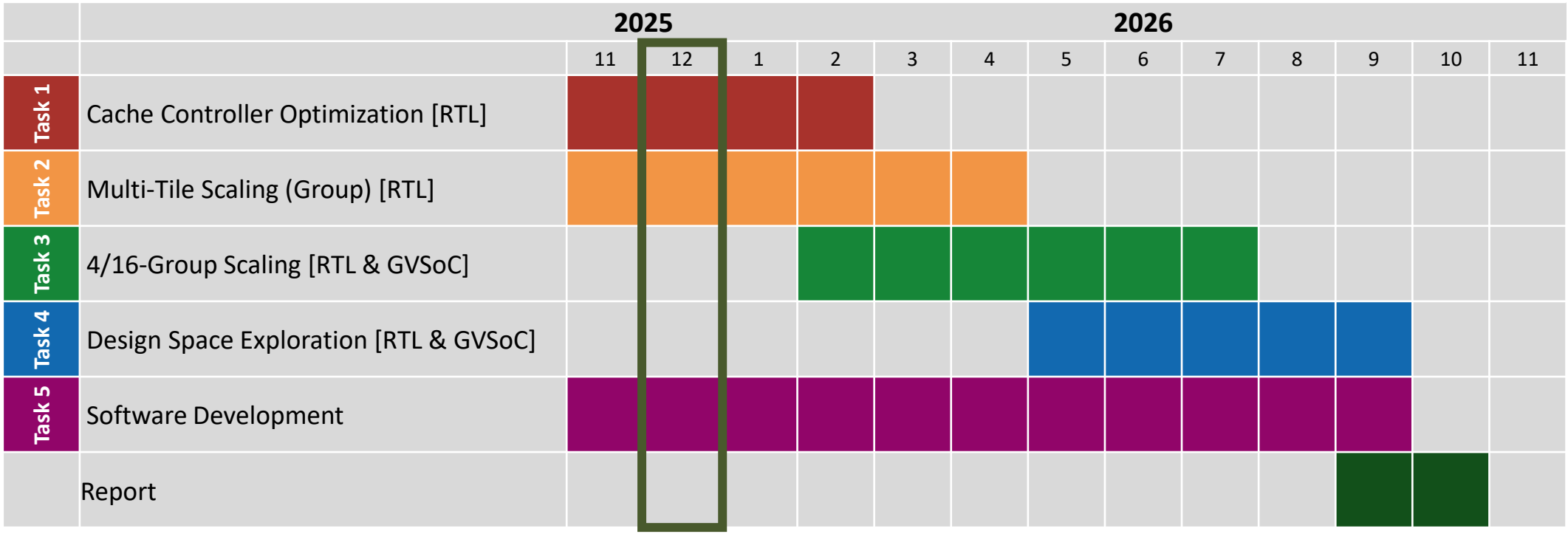
# Hardware Development



# Hardware Development

- **WIP: Multi-Tile Configuration**
  - Able to instantiate multiple tiles in a group
  - Adjusted the location of peripherals
  - Now still working on cross-tile interconnections

# Timeline


2025
2026
11
12
1
2
3
4
5
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7
8
9
10
11
Task 1
Cache Controller Optimization [RTL]
Task 2
Multi-Tile Scaling (Group) [RTL]
Task 3
4/16-Group Scaling [RTL & GVSoC]
Task 4
Design Space Exploration [RTL & GVSoC]
Task 5
Software Development
Report

# Thank you!